Avinash. M
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SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I Avinash. M (192521196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

37/40

Experiments

QUESTIONS:4 **Total Marks:40**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO3 – AT_1

Experiments

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Course : Computer Networks
Code : CSA0707
Date : 20/08/2026

Q1) capture Packets during a TCP connection step

i) AIM:

To capture and analyze the TCP three-way handshake using wireshark and understant how the SUM, SYN,-ACK, and ACK Packets establish a relaible TCP connections.

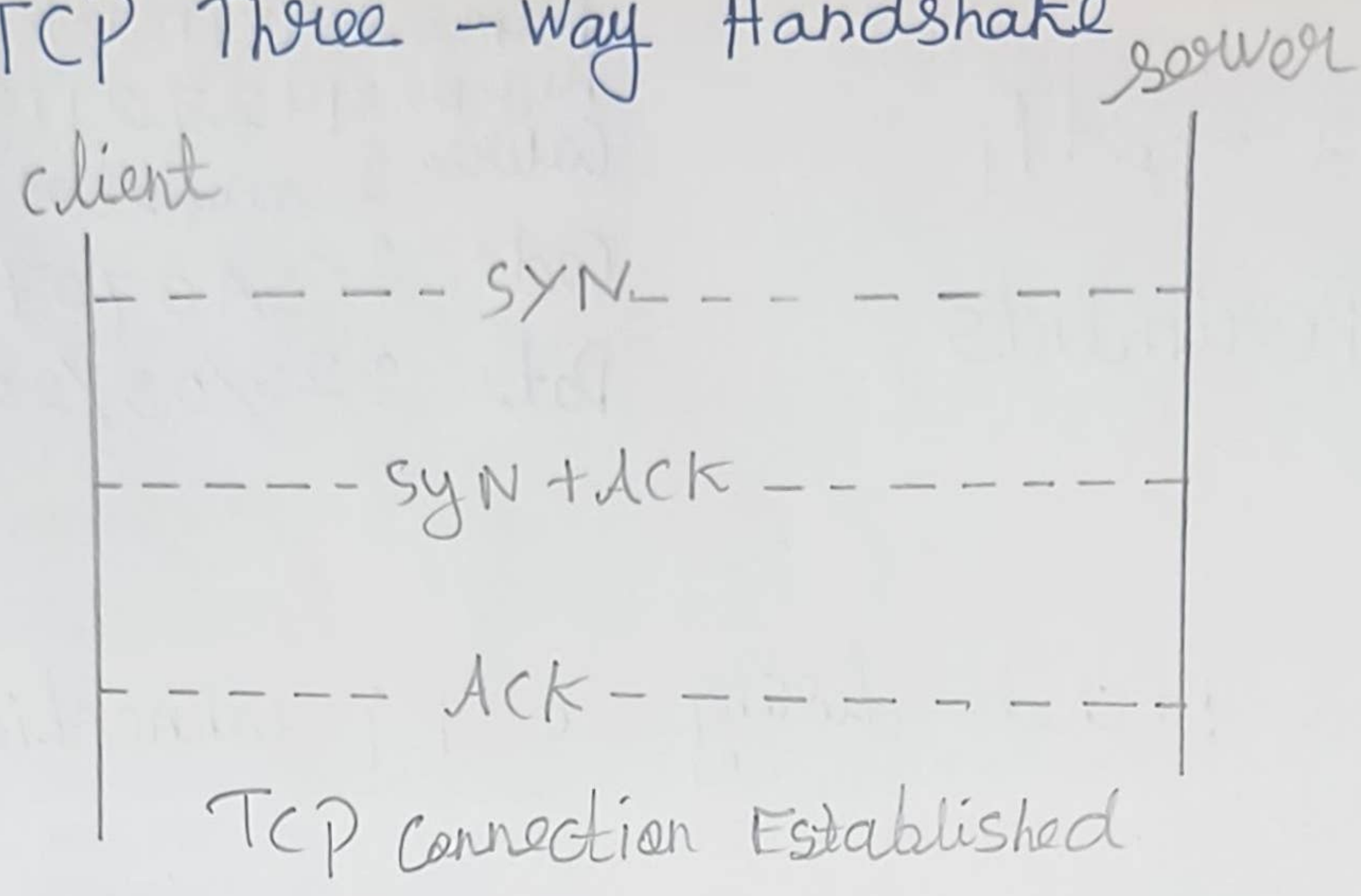
ii) REQUIREMENT:-

*) computer/laptop
*) Internet connection
*) Wirshark
*) Web browser
*) TCP protocal

iii) Procedure:-

*) open Wirshark
*) click start capturing method.
*) open a web browser and accer any website
*) Stop the capture after the connection is established
*) Apply the following display filtered.
*) Identify the TCP Packets involved in connection.

4) TCP Three - Way Handshake



V) How Reliabiality is Established

TCP uses sequence numbers and provide reliable communication. During the handshake both side synchroize their sequence numbers.

Vi) Result :-

The TCP three way handshake was successfully identifed using wireshake, The SYN - ACK, SYN, and Ack packets demostrated how TCP established a connection.

2) Capture a DNS query using UDP in wireshark

i) Aim:

To capture a DNS query using UDD in Wireshark compare the UDP header size with TCP header, follwsing on header size.

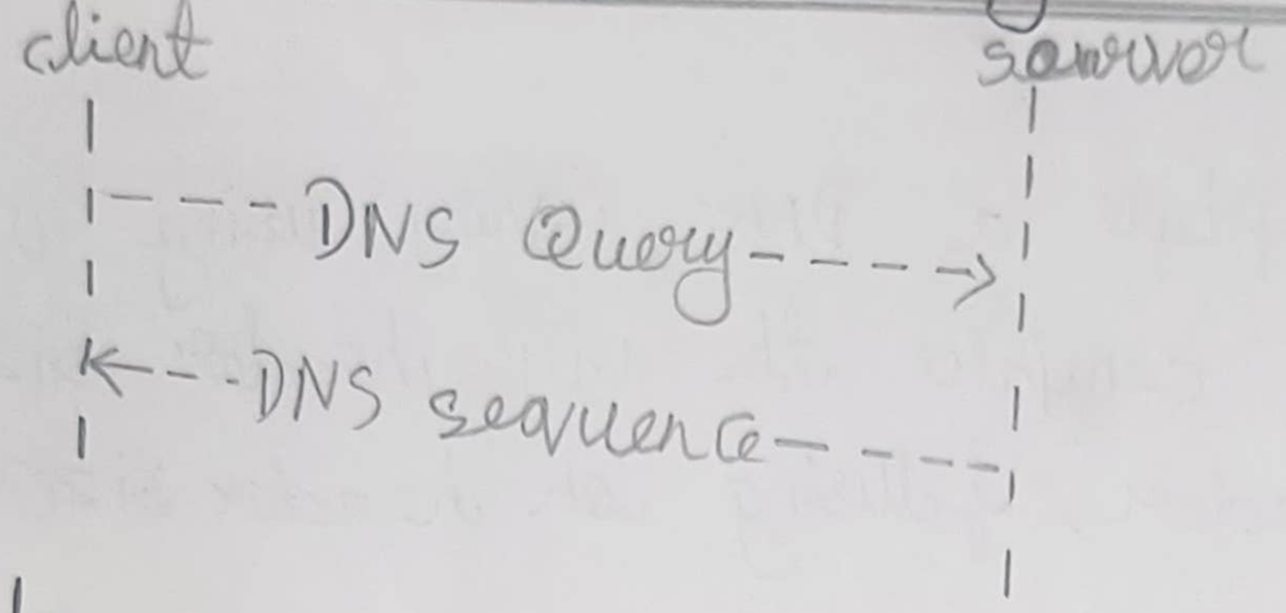
ii) Requirement:

*) Computer/laptop
*) InterConnection
*) Wireshark
*) Web browser
*) DNS services
*) UDP protocal

iii) Procedure:-

i) open wire shark
ii) select the active network interface such as WIFI or Ethernet.
iii) start Packet capture
iv) open a web browser
v) Enter a domain such as google.com
vi) stop. the program
vii) Apply the following Wire shark filter: DNS
viii) select a DNS query Packet.
xi) Expand the source of user datagram.

iv) DNS communication using UDP :-



v) Absence of sequence number & Acknowledgment

unlike TCP, UDP does not contain sequence numbers or acknowledgement number fields in its header Therefore, UDP does not itself confirm whether a Packet has reached.

vi) Result :

The DNS query was successfully captured using Wireshark. The UDP header was observed to be 8 bytes compared with minimum 20 - byte TCP header.

3) simulate Packet loss and observe TCP Retransmission VS UDP :-

i) Aim:

To simulate Packet loss and analyses the behaviour of TCP and UDP using wireshark, especially how TCP retransmits lost Packets while UDP.

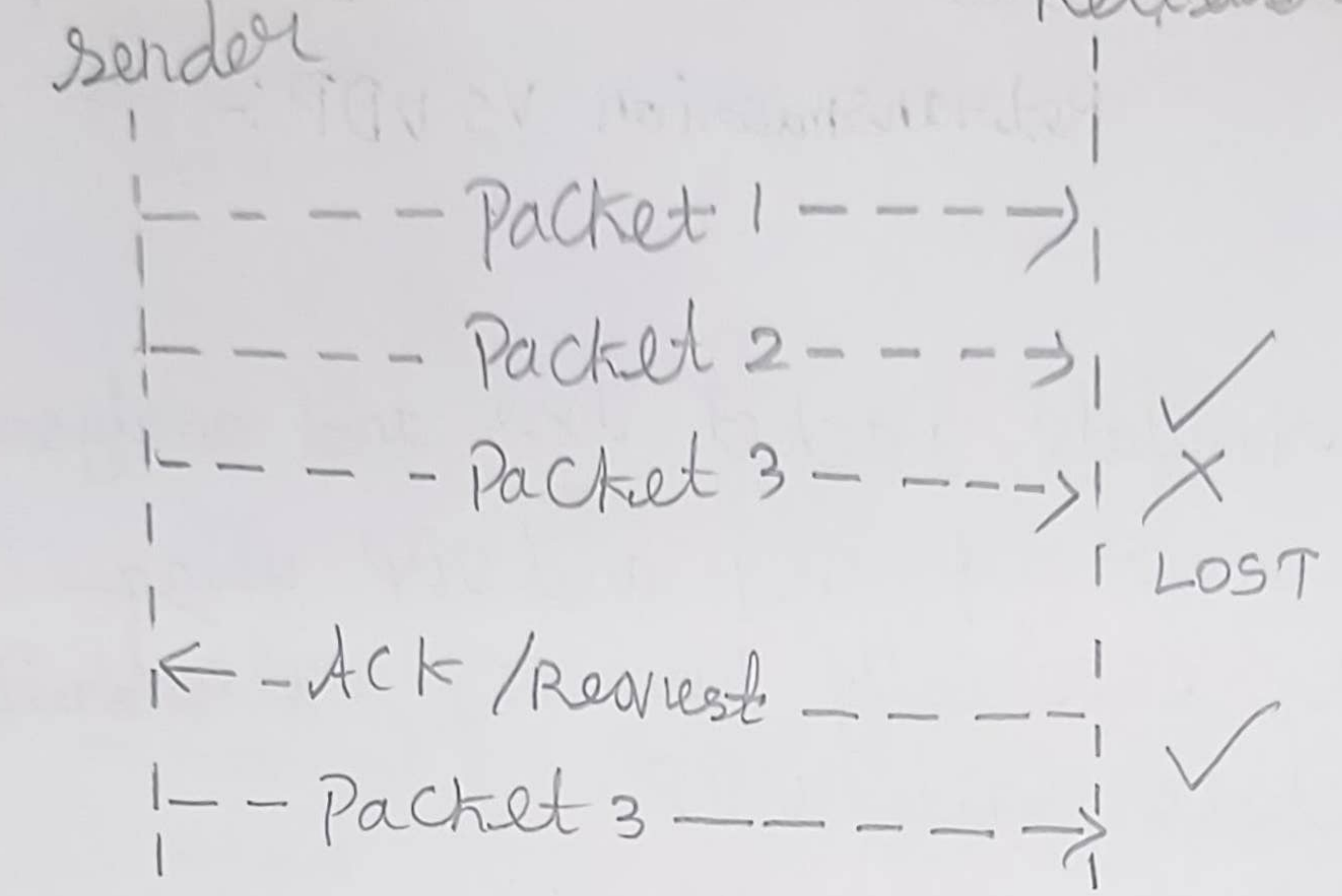
ii) Requirements:-

*) Computer / laptop
*) wireshark
*) Internet / network connection
*) TCP - based communication
*) UDP - based communication

iii) PROCEDURE:

*) open wire shark
*) select the active network interface such as WIFI (Ethernet)
*) start capturing Packets
*) Reconnect the network.
*) Stop the Wireshark capture
*) Examine the TCP Packets before & after the interruption

TCP:-



Result:-

The Packet -loss Experiment that TCP retransmits lost Packete Pa / Where as UDP does not Provide TCP-style transmision. TCP is therefore more retaible but has greater protacal ourhead and can be slower

24). DNS Traffic capture and Resolution Time analy

i) **Aim :**

To capture Dns traffic using wireshake, identily, the DNS query type and response. measures the DNS resolution time.

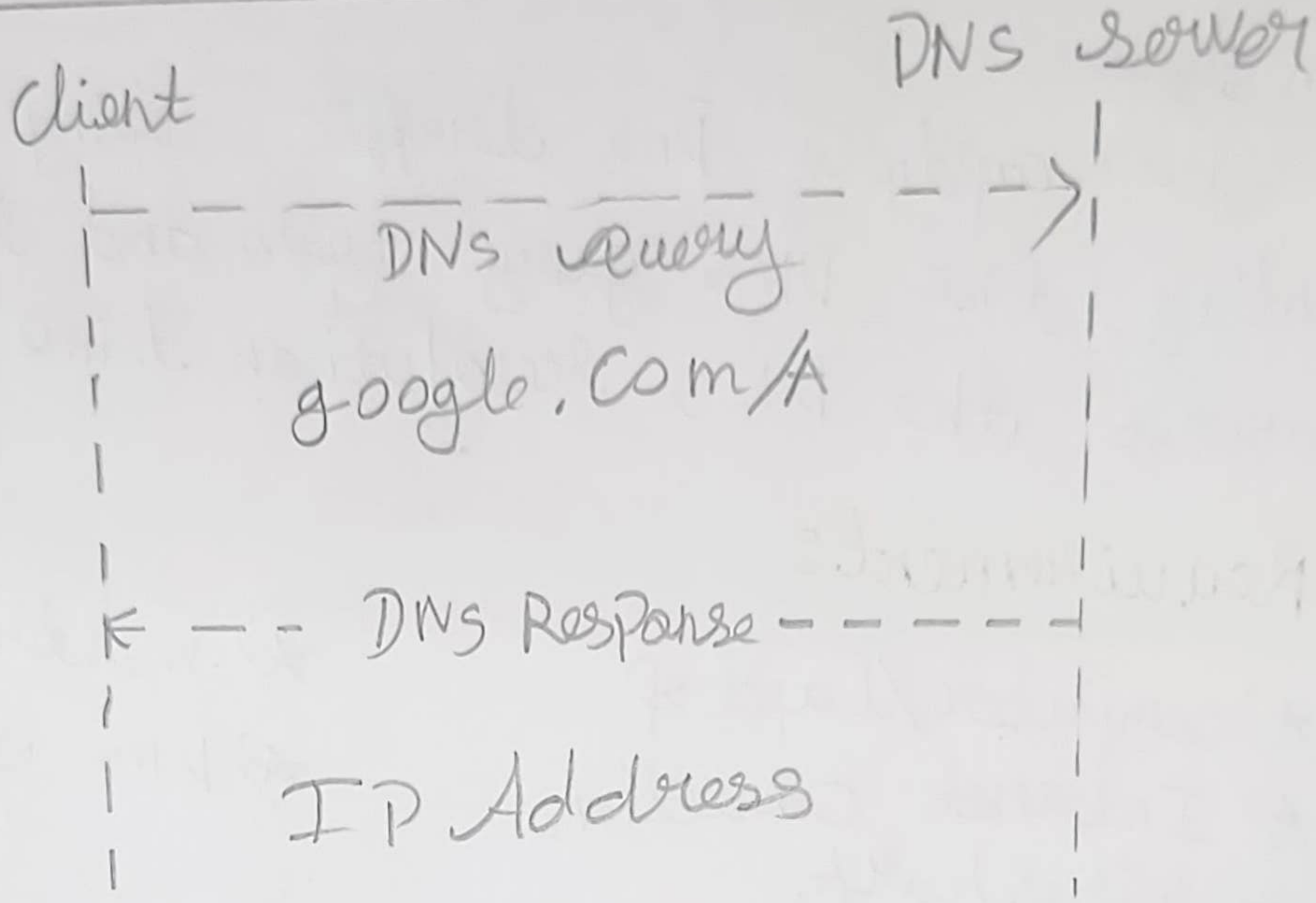
ii) **Requirement :**

*) computer/laptop
*) Internet connection
*) wireshark
*) webbrowser
*) DNs services.

iii) **PROCEDURE :**

*) ope wirshark
*) select the active network interface
*) ~~select packet capture~~
*) open a web browser.
*) stop the Wireshark fiter : DNS
*) Apply the wirchark filter : DNS
*) Identify the Dns query Packet
*) Expand the domain name system (Dns) section.

iv) Dns Query And Response:



v) Why DNS Typialy uses UDP

DNS commonly uses UDP because DNS queries and response are usually small and do not require a TCP

vi) Result:

The Dns trafic was successfully captured using Wireshark. The Dns query type such as AAA AAAA or MX and its corresponding response.